

Chapter 2 — Why OpenFiat Exists

2.1 Understanding Money

Money is one of humanity's oldest and most important inventions.

Every day, billions of people exchange goods and services. A farmer sells vegetables, a software developer builds applications, a teacher educates students, and a mechanic repairs vehicles. Rather than trading goods directly with one another, society uses money as a common medium of exchange.

Money performs three essential functions:

1. It acts as a medium of exchange, allowing people to buy and sell goods and services without needing to barter.
2. It serves as a store of value, allowing wealth to be saved and used in the future.
3. It provides a unit of account, allowing prices to be measured consistently.

Modern economies rely on these three properties every day.

Throughout history, societies have used many different forms of money, including shells, livestock, precious metals, paper banknotes, and electronic bank balances. Although the form of money has changed, its purpose has remained the same: enabling efficient trade.

2.2 What Is Fiat Currency?

Most people today use what is known as *fiat currency*.

Fiat currency is government-issued money that has value because it is recognized and accepted by the issuing country. Examples include:

- United States Dollar (USD)
- Euro (EUR)
- Kenyan Shilling (KES)
- Japanese Yen (JPY)
- British Pound Sterling (GBP)

People receive salaries, pay taxes, purchase groceries, and conduct most daily transactions using fiat currency.

Although many people think of fiat currency as physical cash, most modern money exists only as digital records held by banks and payment providers.

When a bank account shows a balance of \$1,000, those funds typically exist as entries within banking systems rather than as physical banknotes stored in a vault.

2.3 The Rise of Digital Assets

Over the last two decades, technology has transformed how value moves around the world.

Email replaced physical letters.

Video calls replaced long-distance telephone conversations.

Online banking reduced the need to visit physical bank branches.

Similarly, blockchain technology introduced a new way of transferring value across the internet.

Unlike traditional banking systems, blockchain networks allow users to own and transfer digital assets directly without requiring a central institution to maintain account balances or approve every transaction.

Ownership is controlled through cryptographic keys rather than usernames and passwords.

This innovation allows anyone with an internet connection to participate in a global financial network.

2.4 What Is Cryptocurrency?

Cryptocurrency is a form of digital asset secured through cryptography and recorded on a blockchain.

Unlike balances held by commercial banks, cryptocurrencies can generally be transferred directly between users without relying on a central payment processor.

Bitcoin introduced the concept of decentralized digital money in 2009.

Since then, thousands of blockchain networks and digital assets have been created, each designed for different purposes.

Some cryptocurrencies aim to function primarily as money.

Others power decentralized applications, smart contracts, governance systems, gaming platforms, or financial infrastructure.

OpenFiat does not attempt to replace these networks.

Instead, it builds upon one of them—Solana.

2.5 Why Stablecoins Exist

One challenge with many cryptocurrencies is price volatility.

The value of a cryptocurrency may change significantly over short periods of time.

For many everyday transactions, this level of price fluctuation is undesirable.

Stablecoins were created to solve this problem.

A stablecoin is a digital asset designed to maintain a relatively stable value by referencing another asset, most commonly a fiat currency such as the United States Dollar.

For example, one unit of a U.S. Dollar stablecoin is intended to remain approximately equal to one U.S. Dollar.

This stability makes stablecoins useful for:

- Saving value digitally.
- Sending international payments.
- Online commerce.
- Payroll.
- Business settlement.
- Cross-border transfers.

Stablecoins combine many advantages of blockchain technology with the price stability that users expect from traditional currencies.

2.6 Why People Exchange Stablecoins for Fiat

Although stablecoins are increasingly accepted around the world, most people still need local fiat currency for daily life.

Employees receive salaries in local currency.

Businesses pay suppliers using bank transfers.

Governments collect taxes in national currencies.

Utility bills, rent, transportation, and groceries are usually paid using local payment systems.

As a result, users frequently need to exchange between stablecoins and local fiat currency.

Examples include:

- A freelancer receives payment in USDC and converts it to Kenyan Shillings to pay rent.
- A business converts local currency into stablecoins to pay an overseas supplier.
- A traveler exchanges stablecoins for local currency while visiting another country.
- A family member sends stablecoins internationally, and the recipient converts them into local cash.

These exchanges occur millions of times every day.

2.7 What Is a Peer-to-Peer Marketplace?

A peer-to-peer (P2P) marketplace connects individual buyers and sellers directly.

Instead of purchasing assets from the platform itself, users trade with one another.

For example, a merchant may advertise:

"Selling 5,000 USDC. Payment accepted through local bank transfer."

Another user can accept that offer and complete the trade directly with the merchant.

The marketplace facilitates discovery, communication, reputation, and escrow, but it does not normally act as the buyer or seller.

This model allows payment methods to vary between countries and enables participants to choose the merchants, prices, and payment options that best meet their needs.

2.8 The Limitations of Today's P2P Exchanges

Most existing P2P marketplaces provide valuable services.

They simplify trading, reduce fraud through escrow, and help users discover reputable merchants.

However, they typically depend upon centralized infrastructure.

This means:

- A single organization operates the servers.
- A single organization stores reputation data.
- A single organization controls advertisements.
- A single organization resolves disputes.
- A single organization determines which regions and payment methods are supported.
- A single organization can suspend users or discontinue the service.

Even when these organizations operate responsibly, users remain dependent upon their continued existence.

This dependency creates a single point of failure for an otherwise decentralized financial ecosystem.

2.9 Why Existing Decentralized Solutions Are Not Enough

Several decentralized exchanges already exist.

Most of these platforms focus on exchanging one cryptocurrency for another using automated liquidity pools or order books.

While these systems are extremely effective for digital asset trading, they generally do not solve the challenges associated with exchanging digital assets for local fiat currency.

Fiat transactions occur outside the blockchain.

They involve bank transfers, mobile money systems, payment applications, cash deposits, and regional payment networks.

These real-world payment methods require communication, reputation, coordination, and dispute resolution—services that extend beyond the capabilities of a blockchain alone.

OpenFiat is designed specifically to address these challenges.

2.10 Why OpenFiat Exists

OpenFiat exists because decentralized asset ownership deserves decentralized commerce.

If users can own digital assets without trusting a centralized institution, they should also be able to discover trading partners, exchange value, build reputation, communicate, resolve disputes, and participate in marketplace governance without relying on a single company.

Rather than operating another centralized exchange, OpenFiat defines an open protocol that anyone can implement.

Anyone can develop compatible applications.

Anyone can operate network infrastructure.

Anyone can build merchant tools.

Anyone can provide notification services.

Anyone can contribute improvements through open governance.

By separating secure asset settlement on Solana from decentralized marketplace coordination through the OpenFiat protocol, OpenFiat aims to create an open marketplace that remains available, transparent, and community-owned regardless of any individual organization.

The goal is not simply to build another exchange.

The goal is to establish a global public infrastructure for decentralized peer-to-peer digital commerce.

2.11 Looking Ahead

Understanding why OpenFiat exists requires understanding both the strengths and the limitations of existing financial systems.

Blockchain technology solved decentralized ownership.

Stablecoins made digital value practical for everyday commerce.

Peer-to-peer marketplaces enabled direct exchange between individuals.

OpenFiat combines these innovations while removing unnecessary central points of control.

The following chapter introduces the core design philosophy that guided every decision made during the development of the OpenFiat protocol and explains the principles that distinguish it from both traditional financial systems and existing cryptocurrency marketplaces.